

Recommendation **ITU-T K.61 (10/2025)**

SERIES K: Protection against interference

**Guidance on measurement and numerical
prediction of electromagnetic fields for
compliance with human exposure limits for
telecommunication installations**

Recommendation ITU-T K.61

Guidance on measurement and numerical prediction of electromagnetic fields for compliance with human exposure limits for telecommunication installations

Summary

Recommendation ITU-T K.61 helps telecommunication operators to verify compliance with radio frequency electromagnetic field exposure standards promulgated by local or national authorities. This Recommendation provides guidance on measurement methods that can be used to achieve a compliance assessment. It also provides guidance on selecting numerical methods suitable for exposure prediction in various situations.

History^{*}

Edition	Recommendation	Approval	Study Group	Unique ID
1.0	ITU-T K.61	2003-09-06	5	11.1002/1000/6505
2.0	ITU-T K.61	2008-02-29	5	11.1002/1000/9139
3.0	ITU-T K.61	2018-01-13	5	11.1002/1000/13447
4.0	ITU-T K.61	2025-10-22	5	11.1002/1000/16424

Keywords

Electromagnetic fields (EMF), EMF calculations, EMF measurements, exposure assessment.

^{*} To access the Recommendation, type the URL <https://handle.itu.int/> in the address field of your web browser, followed by the Recommendation's unique ID.

FOREWORD

The International Telecommunication Union (ITU) is the United Nations specialized agency in the field of telecommunications, and information and communication technologies (ICTs). The ITU Telecommunication Standardization Sector (ITU-T) is a permanent organ of ITU. ITU-T is responsible for studying technical, operating and tariff questions and issuing Recommendations on them with a view to standardizing telecommunications on a worldwide basis.

The World Telecommunication Standardization Assembly (WTSA), which meets every four years, establishes the topics for study by the ITU-T study groups which, in turn, produce Recommendations on these topics.

The approval of ITU-T Recommendations is covered by the procedure laid down in WTSA Resolution 1.

In some areas of information technology which fall within ITU-T's purview, the necessary standards are prepared on a collaborative basis with ISO and IEC.

NOTE

In this Recommendation, the expression "Administration" is used for conciseness to indicate both a telecommunication administration and a recognized operating agency.

Compliance with this Recommendation is voluntary. However, the Recommendation may contain certain mandatory provisions (to ensure, e.g., interoperability or applicability) and compliance with the Recommendation is achieved when all of these mandatory provisions are met. The words "shall" or some other obligatory language such as "must" and the negative equivalents are used to express requirements. The use of such words does not suggest that compliance with the Recommendation is required of any party.

INTELLECTUAL PROPERTY RIGHTS

ITU draws attention to the possibility that the practice or implementation of this Recommendation may involve the use of a claimed Intellectual Property Right. ITU takes no position concerning the evidence, validity or applicability of claimed Intellectual Property Rights, whether asserted by ITU members or others outside of the Recommendation development process.

As of the date of approval of this Recommendation, ITU had not received notice of intellectual property, protected by patents/software copyrights, which may be required to implement this Recommendation. However, implementers are cautioned that this may not represent the latest information and are therefore strongly urged to consult the appropriate ITU-T databases available via the ITU-T website at <https://www.itu.int/ITU-T/ipr/>.

© ITU 2025

All rights reserved. No part of this publication may be reproduced, by any means whatsoever, without the prior written permission of ITU.

Table of Contents

	Page
1 Scope.....	1
2 References.....	1
3 Definitions	2
3.1 Terms defined elsewhere	2
3.2 Terms defined in this Recommendation.....	2
4 Abbreviations and acronyms	2
5 Conventions	3
6 General principles	3
6.1 Quantities being measured	4
6.2 Typical situations.....	5
7 Technical considerations	5
7.1 Averaging	5
7.2 Quantities.....	6
7.3 Field regions	6
7.4 Shadowing and scattering.....	7
7.5 Variability of the source	8
8 Measurements	8
8.1 Measurement instrumentation	8
8.2 Evaluation of measurement uncertainties.....	11
8.3 Probe selection.....	11
8.4 Procedures	12
8.5 Safety precautions	12
8.6 Field region.....	12
8.7 Multiple sources	12
8.8 Time and spatial variability	12
9 Compliance with the limit: measurement results processing	13
9.1 Identification of individual sources	13
9.2 Intermittent sources	14
9.3 Base stations for mobile communication systems.....	14
Appendix I – Calculation methods.....	15
I.1 General	15
I.2 Methods description	15
I.3 Other near-field models	16
I.4 Practical problems	17
Appendix II – Identification and measurement of the main source of radiation	18
II.1 Treatment of non-conformities – Exceeding the exposure limits	18
II.2 Confirmation measurement	18
Bibliography.....	19

Introduction

This Recommendation helps telecommunication operators to verify compliance with radio frequency electromagnetic field (RF-EMF) exposure standards promulgated by local or national authorities. Guidance on the need to perform an exposure assessment for a telecommunication installation, such as a base station, is provided in [ITU-T K.52]. Indications on the process for putting a telecommunication installation, such as a base station into operation, are provided in [ITU-T K.100]. The assessment is based on the evaluation of the RF-EMF exposure and on accessibility considerations. The RF-EMF exposure evaluation can be carried out by measurement or numerical prediction.

This Recommendation defines tools, methods and procedures that can be used to achieve a compliance assessment. The compliance with RF-EMF exposure standards can be achieved by measurement of electromagnetic field strength, provided that calibrated instruments are used and measurement uncertainty is correctly expressed.

Recommendation ITU-T K.61

Guidance on measurement and numerical prediction of electromagnetic fields for compliance with human exposure limits for telecommunication installations

1 Scope

This Recommendation deals with measurements used for radio frequency electromagnetic field (RF-EMF) strength evaluation to verify that human exposure limits are not exceeded by EMFs produced by telecommunication installations, such as base stations as defined in [IEC 62232], in the frequency range 9 kHz to 300 GHz. This Recommendation also provides guidance on computational methods that can be used to achieve a compliance assessment.

Contact with current exposure due to conductive objects irradiated by an electromagnetic field is not covered in this Recommendation.

The exposure due to the use of mobile handsets or other radiating devices used in close proximity to the human body is not covered. Also, the exposure due to the use of cordless telephone systems and stationary sets intended for the use in wireless telecommunication networks such as digital enhanced cordless telecommunications (DECT), wireless local area network (WLAN), Bluetooth, etc. is not covered.

2 References

The following ITU-T Recommendations and other references contain provisions which, through reference in this text, constitute provisions of this Recommendation. At the time of publication, the editions indicated were valid. All Recommendations and other references are subject to revision; users of this Recommendation are therefore encouraged to investigate the possibility of applying the most recent edition of the Recommendations and other references listed below. A list of the currently valid ITU-T Recommendations is regularly published. The reference to a document within this Recommendation does not give it, as a stand-alone document, the status of a Recommendation.

- [[ITU-T K.52](#)] Recommendation ITU-T K.52 (2024), *Guidance on complying with limits for human exposure to electromagnetic fields*.
- [[ITU-T K.70](#)] Recommendation ITU-T K.70 (2020), *Mitigation techniques to limit human exposure to EMFs in the vicinity of radiocommunication stations*.
- [[ITU-T K.100](#)] Recommendation ITU-T K.100 (2024), *Measurement of radio frequency electromagnetic fields to determine compliance with human exposure limits when a base station is put into operation*.
- [ITU-R BS.1698] Recommendation ITU-R BS.1698 (2005), *Evaluating fields from terrestrial broadcasting transmitting systems operating in any frequency band for assessing exposure to non-ionizing radiation*.
- [IEC 62232] IEC 62232 (2025), *Determination of RF field strength, power density and SAR in the vicinity of base stations for the purpose of evaluating human exposure*.
- [IEC 62479] IEC 62479 (2010), *Assessment of the compliance of low-power electronic and electrical equipment with the basic restrictions related to human exposure to electromagnetic fields (10 MHz to 300 GHz)*.

3 Definitions

3.1 Terms defined elsewhere

This Recommendation uses the following terms defined elsewhere:

3.1.1 far field region [b-IEC 60050 712]: That region of the electromagnetic field of an antenna wherein the predominant components of the field are those which represent a propagation of energy and wherein the angular field distribution is essentially independent of the distance from the antenna.

NOTE 1 – In the far field region, all the components of the electromagnetic field decrease in inverse proportion to the distance from the antenna.

NOTE 2 – For a broadside antenna having a maximum overall dimension D which is large compared to the wavelength λ , the far field region is commonly taken to exist at distances greater than $2D^2/\lambda$, from the antenna in the direction of maximum radiation.

3.1.2 radiating near field (region) [b-IEC 60050 712]: That region of space between the reactive near field region and the far field region, wherein the predominant components of the electromagnetic field are those which represent a propagation of energy, and wherein the angular field distribution is dependent upon the distance from the antenna.

NOTE – If the antenna has a maximum overall dimension which is not large compared to the wavelength, the radiating near field region may not be identifiable in practice.

3.1.3 wavelength [b-IEC 60050 103]: Distance, in the direction of propagation of a sinusoidal wave, between two successive points where the phases of the characteristic quantity differ by 2π radians.

3.2 Terms defined in this Recommendation

This Recommendation defines the following terms:

3.2.1 actual maximum (max) approach: Evaluation method of radio frequency electromagnetic field (RF-EMF) exposure taking into account the actual maximum thereof.

NOTE – For further information, see [IEC 62232], [ITU-T K.52], [ITU-T K.100].

3.2.2 radio frequency (RF): Any frequency at which electromagnetic radiation is useful for telecommunication.

NOTE – In this Recommendation, radio frequency refers to the frequency range 9 kHz-300 GHz allocated by ITU-R Radio Regulations.

3.2.3 specific absorption (SA): The quotient of the incremental energy (dW) absorbed by (dissipated in) an incremental mass (dm) contained in a volume element (dV) of a given density (ρ_m).

NOTE – The specific absorption is expressed in units of joules per kilogram (J/kg).

3.2.4 specific absorption rate (SAR): The time derivative of the incremental energy (dW) absorbed by (dissipated in) an incremental mass (dm) contained in a volume element (dV) of a given mass density (ρ_m).

4 Abbreviations and acronyms

This Recommendation uses the following abbreviations and acronyms:

AF	Antenna Factor
APC	Automatic Power Control
APD	Absorbed Power Density
CF	Calibration Factor
DECT	Digital Enhanced Cordless Telecommunications

EIRP	Equivalent Isotropically Radiated Power
EM	Electromagnetic
EMF	Electromagnetic Field
FDTP	Finite-Difference Time-Domain
MOM	Method of Moments
MR	Multiple-Region
NEC	Numeric Electromagnetic Code
PC	Personal Computer
RF	Radio Frequency
RMS	Root Mean Square
SA	Specific Absorption
SAR	Specific Absorption Rate
WLAN	Wireless Local Area Network

5 Conventions

None.

6 General principles

A procedure for achieving compliance with radio frequency electromagnetic field (RF-EMF) exposure limits is provided in [ITU-T K.52]. The steps needed to achieve compliance are:

- 1) Identify appropriate compliance limits.
- 2) Determine whether an electromagnetic field (EMF) exposure assessment for the installation of equipment in question is needed.
- 3) If an EMF exposure assessment is needed, it may be performed by calculations or measurement.
- 4) If the EMF exposure assessment indicates that pertinent exposure limits may be exceeded in areas where people may be present, mitigation/avoidance measures should be applied.

Additional information on RF-EMF exposure assessments can be found in [IEC 62232] and [IEC 62479], as well as [ITU-T K.70], [ITU-R BS.1698] and [b-CSN EN 50413]. It is also recommended to ensure that exposure assessment is conducted in accordance with the applicable national or regional standards and regulations.

As defined in [ITU-T K.100], telecommunication installations with some predefined criteria may be put into operation without proceeding with detailed measurements or calculations. The more sophisticated assessment methods defined in this Recommendation are used to refine the zone boundaries obtained using [ITU-T K.52] or for complex situations where the methods of [ITU-T K.52] may be insufficient. For example, it may be useful to refine the results of [ITU-T K.52] where they indicate the appearance of an exceedance zone or occupational zone marginally. A measurement or more accurate calculation can help determine whether the zone determination is correct or is an artefact of the conservative estimation methods of [ITU-T K.52]. Another example where measurements may be needed is in complex scattering environments or for environments with a number of significant sources of electromagnetic (EM) radiation.

To ensure accurate results and avoid overestimations, RF-EMF exposure assessments should be performed using the actual maximum transmitted power or equivalent isotropically radiated power

(EIRP) of the radio transmitters. The actual maximum approach is described in [IEC 62232], [ITU-T K.52] and [ITU-T K.100].

6.1 Quantities being measured

Most documents provide RF-EMF exposure limits in terms of basic restrictions and reference (or derived) levels. The basic restrictions address the fundamental quantities that determine the physiological response of the human body to EMFs. Basic restrictions apply to situations with the body present in the field. The basic restrictions for human exposure are expressed as the specific absorption rate (SAR), specific absorption (SA), absorbed power density (APD) and current density. SAR, SA, APD and current density assessments are not within the scope of this Recommendation; more information can be found in [IEC 62232] and [b-IEC PAS 63446].

As the basic restrictions are difficult to measure directly, most RF-EMF exposure standards and recommendations provide derived reference levels for electric field strength, magnetic field strength and power density. The reference levels apply to situations where the assessment of RF-EMF is not influenced by the presence of a body. The normative part of this Recommendation provides guidelines for the measurement of reference levels.

Reference levels may be exceeded if the exposure condition can be shown to produce SAR, SA, APD and induced current density below the basic restrictions. Therefore, Appendix I provides guidance on selecting computational procedures that can be used to calculate these metrics.

$$SAR = \frac{d}{dt} \frac{dW}{dm} = \frac{d}{dr} \left(\frac{1}{\rho m} \frac{dW}{dV} \right)$$

SAR is expressed in units of watts per kilogram (W/kg).

SAR can be calculated by:

$$SAR = \frac{\sigma E^2}{\rho m}$$

$$SAR = c \frac{dT}{dt}$$

$$SAR = \frac{J^2}{\rho m^\sigma}$$

where:

E is the value of the electric field strength in body tissue in V/m

σ is the conductivity of body tissue in S/m

ρm is the density of body tissue in kg/m³

c is the heat capacity of body tissue in J/kg°C

$\frac{dT}{dt}$ is the time derivative of temperature in body tissue in °C/s

J is the value of the induced current density in the body tissue in A/m²

6.2 Typical situations

The measurement problem typically approaches one of the following cases:

- 1) The source of the RF-EMF and at least some of its characteristics are known. The RF-EMF from other sources is negligible for compliance considerations. The objective is to determine the compliance zones for this known source.

- 2) The sources of the RF-EMF are not known. The objective is to determine compliance in a particular location, or to survey the EM fields in the out-of-band region to confirm that other EM sources can be neglected.
- 3) The objective is to determine compliance in a particular location, and if non-compliance is found, to determine the relative contribution of the sources to the non-compliance.

In case 1, the emission frequency band should be known precisely. The transmitted power, polarization and the antenna pattern may be known approximately. Thus, the measurements can focus on to the frequency range of interest. [ITU-T K.52] or [IEC 62232] should be used to obtain an estimate of the field strength in order to determine appropriate instrumentation.

In case 2, a survey of the entire frequency spectrum may be required. An alternative is measurement with a wideband probe that integrates various frequencies. Case 3 is an extension of case 2. If the initial measurement indicates non-compliance, frequency-selective measurements, an antenna and spectrum analyser, for example, are needed.

Additional guidelines on how to conduct an RF-EMF exposure assessment and detailed descriptions of measurement and calculation methodologies can be found in [IEC 62232].

7 Technical considerations

7.1 Averaging

7.1.1 Time averaging

Limits are usually expressed as root mean square (RMS) values of a continuous wave averaged over a defined period. For example, the basic restrictions and reference levels specified in [b-ICNIRP] are to be averaged over any 6-minute period for local exposure and over any 30-minute period for whole-body exposure. Therefore, for time-dependant signals, an elaboration of measurement results (post-processing procedure) may be necessary for comparison with the limit.

7.1.2 Spatial averaging

The basic restrictions typically comprise two categories: local limits and whole-body limits. The local limits are pertinent to localized exposures due to small radiators close to the body, such as mobile phones, and are to be averaged over a small mass or surface. The whole-body limits are to be averaged over the whole body and are applicable in the case of RF-EMF exposure from base stations in mobile networks and other telecommunication installations.

The whole-body reference levels are generally intended to be spatially averaged values over the dimensions of the body of the exposed individual, but with the important proviso that the local basic restrictions or reference levels on exposure are not exceeded.

For telecommunication installations, the highest field values occur close to the antennas in regions where the fields may vary appreciably on the scale of the size of the human body. In these cases, spatial averaging can be used to obtain a more accurate result.

7.2 Quantities

Exposure standards usually refer to reference levels expressed as electric and magnetic field strengths or power density. They are individually measured only when it is required by the field properties related to the field regions.

7.3 Field regions

The properties of EMFs need to be taken into consideration for their measurement and evaluation. For example:

- measurement of both the electric and magnetic components may be necessary in the non-radiating near-field region;
- for numerical prediction, the far-field model usually leads to an overestimation of the field if applied in near-field regions.

Therefore, it is important to be aware of the boundaries of each field region before starting a compliance procedure. Information about field regions is provided in [IEC 62232].

7.3.1 Reactive near-field zone

The reactive near-field zone is the portion of the near-field region that is immediately surrounding the antenna and where the reactive field predominates. This region is commonly assumed to extend to a distance of one wavelength from the antenna.

7.3.2 Reactive-radiative near-field region

At the boundary to the reactive near-field zone, a transition region may be defined wherein the radiating field is beginning to be important compared with the reactive component. This outer region extends to a few (e.g., 3λ) wavelengths from the electromagnetic source.

7.3.3 Radiating near-field (Fresnel) zone

The region of the field of an antenna between the reactive near-field and the far-field region, wherein the radiation field predominates. Although the radiation is not propagating as a plane wave, the electric and magnetic components can be considered locally normal; moreover, the ratio E/H can be assumed constant (and almost equal to Z_0 , the intrinsic impedance of free space). This region exists only if the maximum dimension D of the antenna is large compared with the wavelength λ .

7.3.4 Radiating far-field zone

The region of the field where the angular field distribution is essentially independent of the distance from the antenna and the radiated power density [W/m^2] is constant. The inner boundary of the radiating far-field region is defined by the larger value between 3λ and $2D^2/\lambda$ (i.e., the limit is $2D^2/\lambda$ if the maximum dimension D of the antenna is large compared with the wavelength λ). In the far-field region, the field components are transverse and propagate as a plane wave.

The above regions are shown in Figure 1 (where D is supposed to be large compared with the wavelength λ) and the main properties of an electromagnetic field (EMF) in different field regions are shown in Table 1.

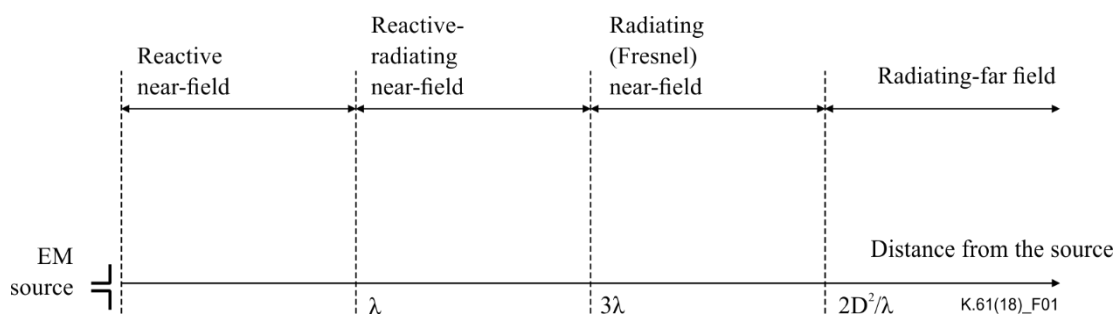


Figure 1 – Field regions around an EM source (the antenna maximum dimension D is supposed to be large compared with the wavelength λ)

Table 1 – Main properties of electromagnetic field in different field regions

	Reactive near-field	Reactive-radiating near-field	Radiating near-field	Radiating far-field
Inner boundary	0	λ	3λ	$\text{Max}(3\lambda; 2D^2/\lambda)$
Outer boundary	λ	3λ	$\text{Max}(3\lambda; 2D^2/\lambda)$	∞
Power density S [W/m ²]	$S \leq E H $	$S \leq E H $	$S \leq E H $ $= \frac{ E ^2}{Z_0} = Z_0 H ^2$	$S = E H $ $= \frac{ E ^2}{Z_0} = Z_0 H ^2$
$E \perp H$	No	No	Locally	Yes
$Z = E/H$	$\neq Z_0$	$\neq Z_0$	$\approx Z_0$	$= Z_0$

7.4 Shadowing and scattering

The RF-EMF exposure varies with spatial position due to the effect of reflections and scattering from adjacent conducting structures. The scale of this variability is a function of the wavelength. It is important to consider this variability in order to determine the locations of maximum exposure and to use spatial averaging as appropriate.

Since the exposure standards specify the limits for the exposure of the human body, the effect of the body on the field pattern should be considered. For example, Figure 2 shows a situation where the presence of a human body would absorb part of the incident wave, creating a shadow region that precludes a reflection that would otherwise enhance the field at the position of the body. These types of effects, especially at microwave frequencies, can lead to an overestimation of the field during measurements or numerical calculations near reflecting objects.

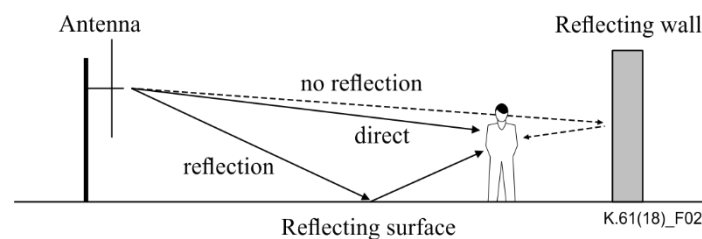


Figure 2 – An illustration of a multi-path alteration due to the presence of a human body

7.5 Variability of the source

Telecommunication sources are sometimes variable. Variability of transmitted power and antenna patterns is especially important. This variability presents a special challenge for measurements, since the exact state of the transmitter at the time of measurement may not be known.

7.5.1 Power variability

The exposure assessment must take into account either the configured maximum total radiated power or the actual maximum power or EIRP from the transmitter. The power transmitted in a telecommunication system could vary due to automatic power control (APC) or channel use

variability. APC adjusts output power to compensate for adverse propagation conditions. Channel variability falls into two categories:

- 1) Dynamic channel allocation, where the channels are turned on or off as needed; or
- 2) Variation in channel occupancy, where the amount of data transmitted over a channel varies; however, even if no data is transmitted, the channel carrier remains. Channel occupancy variation affects the modulation of the signal; however, this effect is expected to be small.

7.5.2 Antenna variability

Certain telecommunication systems use active antennas that can dynamically vary their radiation pattern. The actual maximum approach described in [IEC 62232] takes into account the antenna pattern variability in the spatial domain.

7.5.3 Intermittent sources

Certain sources used in telecommunications are intermittent. Such sources emit RF energy only if they need to transmit information.

Such sources may operate in a regular manner, transmitting data at regular intervals or to a defined schedule.

Such sources may also operate in an irregular manner, transmitting data only if activated by an operator or if a sufficient amount of data has been accumulated to trigger transmission.

8 Measurements

8.1 Measurement instrumentation

8.1.1 Characteristics

The following general characteristics of measurement devices are important in their selection.

8.1.1.1 Frequency range

There are two classes: broadband and narrowband.

- 1) Broadband devices (such as the commonly used electric and magnetic probes) do not give information on the frequency spectrum. Nevertheless, frequency-selective measurements on large bands are possible by using small broadband antennas (e.g., biconical, horn, etc.) or more sophisticated and expensive devices.
- 2) Narrowband devices are generally antennas with flat antenna factors (AFs) over limited spectrum ranges (e.g., dipole antennas) and can be used for frequency-selective measurement.

8.1.1.2 Antenna directivity

The antenna response may be isotropic or directional.

For isotropic devices, the response is expected to be independent of the direction of the incident EMF.

For directional devices, the response is expected to be dependent on the direction of the incident EMF. Directional devices are generally polarized and have axial symmetry in the radiation pattern. Thus, proper orientation of the device in 3 orthogonal axes is necessary for field reconstruction.

8.1.1.3 Quantity measured

The majority of devices measure either the electric field or the magnetic field.

The distinction is important in the reactive field region.

In the far-field region, it is possible to measure either the electric or the magnetic field component and determine the equivalent power density. However, measurement devices for the electric field component are usually preferred. The equivalent power density within the far-field region is obtained from the measured field by calculation, as shown in Table 1.

8.1.2 Device selection

The choice of devices for RF-EMF measurement is determined by certain factors, for instance:

- the existing standards to be complied with (e.g., limits may be frequency-dependent);
- the number and the characteristics of RF-EMF sources; and
- the field regions (i.e., reactive near-field, radiating near-field, and far-field) in which the measurements are made.

The choice of measurement equipment is strongly related to the measurement procedure. The accuracy of measurement results depends on measurement procedures as well as on the characteristics of the measurement instruments used. More information on uncertainty is provided in clause 8.2.

8.1.3 Calibration requirements

8.1.3.1 Calibration factor

For broadband probes, the calibration factor (CF) is defined by the following formula:

$$CF = \frac{E_{ref}}{E_{meas}}$$

It is the ratio between the expected electric reference field strength (E_{ref}) and the value (E_{meas}) read on the personal computer (PC) or on a dedicated receiver unit. This factor is mainly a function of frequency and, in the presence of non-linearity error, of field strength. The CF is determined as a frequency function. For each frequency, the CF value shall be known with an uncertainty of less than 1 dB. Errors due to frequency interpolation are included in the tolerable uncertainty of CF.

8.1.3.2 Antenna factor

The antenna factor (AF) is defined for antennas and frequency-selective probes as the ratio:

$$AF = \frac{E_{ref}}{V} [m^{-1}]$$

where E_{ref} [V/m] is the electric field strength at the probe and V [V] is the voltage measured by the spectrum analyser. This factor is primarily a function of frequency but, in the presence of non-linearity error, it may depend on field strength, as well. The AF is determined as a frequency function. For each frequency, the AF value shall be known with an expanded uncertainty (i.e., 95% statistical confidence) of less than 2 dB. The maximum tolerable uncertainty also includes the error due to frequency interpolation (when needed).

8.1.3.3 Isotropy

An isotropic probe is almost always useful in compliance measurements for a telecommunication installation. The isotropic response is usually achieved by a triaxial antenna system, where the three axes are arranged to be mutually orthogonal. The deviation from an ideal isotropic response is measured in the isotropy test. The deviation is called the isotropic error and in general it is a function of the incident wave direction. It can be evaluated:

- by measuring the difference from a cosine response of each axis, if they are clearly spatially identified and a signal from each axis is available; or
- by checking the whole probe response, if it is not possible to clearly define the position of each axis or a single axis signal is not available.

The mean deviation from the isotropic response should be less than 1 dB.

8.1.3.4 Linearity

A linear response with respect to the field amplitude is required: a linearity error would mean that the antenna and the calibration factors are functions of the test field strength. Thus, the linearity test should be the starting point of the whole characterization process of the probe. The test is carried out, over as wide a dynamic range as possible, by verifying the relationship between radiated power and the electric field or voltage measured. The relationship is linear in logarithmic units: the uncertainty band on the linear regression shall have the same magnitude as the measurement uncertainty. If the condition is not fulfilled, a linearity error is probable and the following actions are suggested:

- in the characterization process: CF or AF are measured for different amplitudes of the test wave and different results are obtained;
- in the compliance survey: differences due to field strength can be managed by widening the measurement uncertainty or by considering different factors for different field amplitudes (where it is possible).

Checking the linearity at some frequencies may be useful. The maximum tolerable deviation from a linear response is 1 dB.

8.1.3.5 Pulsed signal

Due to their modulation and their multiple-media access, radio-mobile digital systems have pulsed transmissions. Therefore, when the characterization is carried out with a continuous wave test field, it is necessary to verify whether a pulsed test field introduces any changes in the tested characteristics.

If differences in CF and AF, as determined by a pulsed test wave and a continuous wave, are less than the relevant uncertainties, the measurement instruments can be used without regard to the type of signal they are measuring.

8.1.3.6 Multiple signal integration

Verifying the correct integration of different signals with different frequencies is an important test for non-selective broadband probes. It means verifying that the measurement result is correctly given by an RMS formula:

$$E_{rms} = \sqrt{\sum_i E_i^2}$$

The test can be easily carried out with two RF sources: results shall comply with the condition:

$$20\log_{10}\left\{\frac{\left|E_{mes} - \sqrt{E_1^2 + E_2^2}\right|}{E_{mes}}\right\} < 0.5 \text{ dB}$$

where:

E_{mes} is the measured electric field

E_1 and E_2 are the actual field values

or less than the measurement uncertainty on the electric field or voltage.

8.1.3.7 Axial rejection

The response of an axis irradiated by a cross-polarized incident wave is measured in the test. A low axial rejection could have important effects on the electric field strength measurement when it is determined as the RMS value of three orthogonal components.

8.2 Evaluation of measurement uncertainties

Measurement uncertainties for measurements of fields are the results of errors due to system instrumentation, field probe response and calibration and the extrapolation, interpolation and integration algorithms used to determine the averaged field. For evaluation and expression of uncertainties, see [IEC 62232] and [b-JCGM 100:2008].

The target expanded uncertainty for in situ field measurements is less than or equal to 4 dB, which is considered industry best practice. The expanded uncertainty for the RF exposure evaluation used for in situ RF EMF exposure measurements shall not exceed 6 dB. Under this condition, the evaluation result shall be compared directly to the exposure limit. However, the entity performing the uncertainty analysis also needs to be aware of the applicable regulation. More information can be found in [IEC 62232] or [b-CSN EN 50413].

8.3 Probe selection

8.3.1 Probe size

If measurements in the near field are being made, the dimension of the probe sensor should be less than one wavelength at the highest operating frequency.

8.3.2 Frequency range

General consideration: use broadband wherever possible (as it is simpler and shorter), but often a frequency-selective measurement is required (in general, when it is not possible to distinguish one main source and when the measurement results need an elaboration to be compared with an RMS limit).

Selective measurement is usually necessary in the case of:

- multiple sources with different limits;
- multiple sources to which different measurement techniques are recommended (e.g., post-processing for different mobile technologies or others); or
- when it is necessary to determine the relative contribution of multiple sources.

8.3.3 Directivity

A non-directional probe is preferred.

8.4 Procedures

Before performing a measurement of potentially non-compliant RF-EMF levels, an approximate assessment as described in [ITU-T K.52] should be performed. This will permit an estimate of the expected RF-EMF exposure, the boundaries of the compliance zones and consequently will help in the selection of appropriate test instruments and test procedures.

8.5 Safety precautions

Personnel should take appropriate safety precautions while performing measurements. If measurements are performed in the exceedance zone, precautions specified in [ITU-T K.52] should be followed. Also, precautions against indirect effects, such as contact currents, should be observed.

8.6 Field region

What is to be measured (E or H) depends on where (reactive or radiating field) the observer is and on the field impedance.

- Reactive near-field: measure both the E and H components or evaluate the SAR.

- Reactive radiating near-field: if no information on the field impedance is available, measure both the E and H fields; if information on the field impedance is available, it is possible to measure only one field component, provided that conservative results are obtained:
 - Measure only E component if $\frac{E}{H} > Z_0 = 120 \times \pi [\Omega]$ i.e., high impedance EMF
 - Measure only H component if $\frac{E}{H} < Z_0 = 120 \times \pi [\Omega]$ i.e., low impedance EMF
- Radiating near-field: measure only the E component, the free space impedance (Z_0) is assumed (as differences are small compared with the measurement uncertainties).
- Radiating far-field: measure only the E component.

For exposure at positions located very close to the source, determining the SAR instead of performing a field measurement may be preferable.

8.7 Multiple sources

The effects of multiple sources operating at different frequencies should be considered according to [b-ICNIRP] or the applicable RF-EMF exposure standard, usually in a weighted sum, where each individual source is prorated according to the limit applicable to its frequency.

8.8 Time and spatial variability

Multi-path reflections can create non-uniform field distributions. Therefore, to assess the whole-body human exposure, an averaging process is required. The field values should be determined at N points as described in Figure 3. Three points are basically recommended (Figure 3a), but if accuracy is required the number can be increased to six (Figure 3b), nine (Figure 3c), twenty (Figure 3d), etc., in accordance with national or regional standards and regulations. In all cases, the uncertainty of the evaluation should be determined. More information on spatial averaging can be found in [IEC 62232].

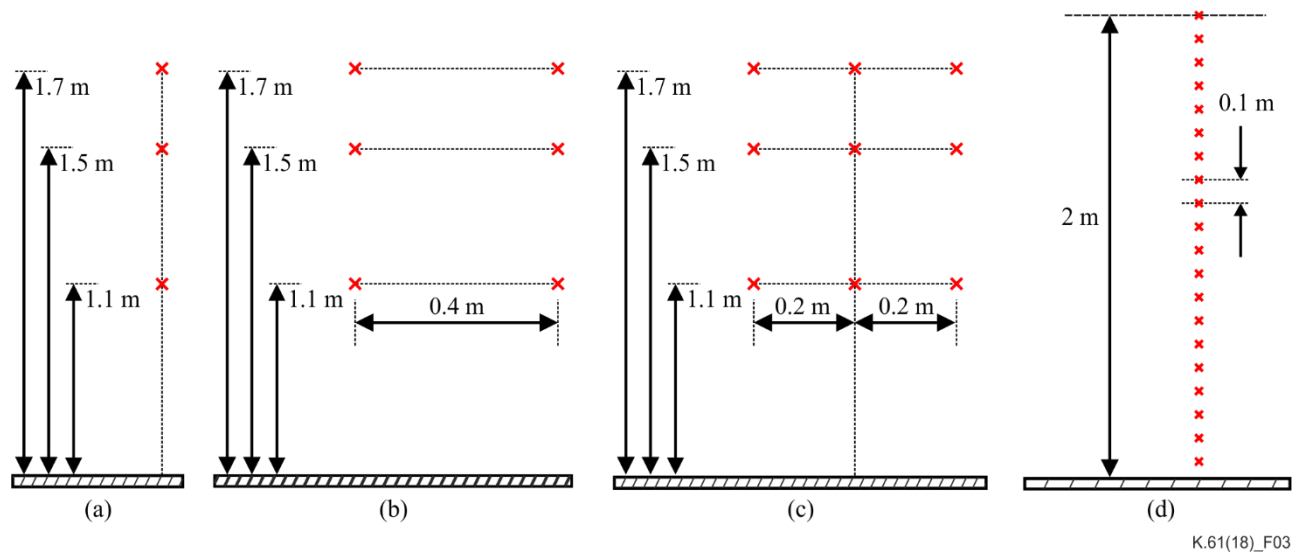


Figure 3 – Measurement points for spatial averaging

The formula for calculating spatially averaged field values is as follows:

$$(E \text{ or } H) = \sqrt{\frac{\sum_{i=1}^N (E_i \text{ or } H_i)^2}{N}}$$

where:

N : is the number of points, i.e., three, six, nine, twenty, etc.

Measurement should not be conducted close to metallic objects to avoid coupling with the probe. For example, it may be sufficient to keep the edge of the probe at least 3 "probe lengths" from the metallic object.

In case of multiple sources, the measurement area should be divided into a grid of about one square metre and measurements should be performed at individual grid points. Large field gradients can exist in the near field of a radiator. Measurements should be performed sufficiently close together to accurately determine the compliance zone boundaries.

In areas of expected time variability of the source, measurements may need to be performed over an extended time period. For example, in case of channel variability, measurements should be made during periods of peak usage.

NOTE – Initial measurements in a grid or near the radiator, as described in this clause, yield the maximum point field values. These values represent the most conservative evaluation of the exposure. It is possible to define the compliance zones based on these conservative values. If a more refined estimation is desired, spatial averaging, as described in clause 7.1.2, should be used.

9 Compliance with the limit: measurement results processing

9.1 Identification of individual sources

The probe used for external field measurements in determining compliance should generally be isotropic, non-directional and non-polarized. Also, the probe should not produce significant scattering of the incident electromagnetic field, and the leads from the sensor to the meter should not interact significantly with the field. However, such a probe cannot differentiate between different sources.

Frequency-selective or directional measurements are needed to identify the contribution of individual sources. For example, a combination of an antenna and a spectrum analyser allows for a more precise measurement of individual frequency, direction and polarization field components. However, this makes the measurements more complicated as it is necessary to measure and sum three polarizations separately. Also, in complex scattering environments, it may be necessary to measure the fields in various directions. It is also possible to use the antenna and spectrum analyser combination to verify the frequency and origin of the emissions measured by the isotropic probe.

In cases when non-conformity with exposure limits is found, the procedure for the identification of the main source of radiation, according to Appendix II, should be applied.

9.2 Intermittent sources

Neither the isotropic wideband probe nor a spectrum analyser can measure the duration of an intermittent source. The field probe measures the maximum (peak) field value, while the spectrum analyser measures the maximum spectral density in the frequency domain. To obtain proper time-averaging, the duration of an intermittent transmission has to be determined from the operational requirements of the system.

9.3 Base stations for mobile communication systems

The preferred method for RF-EMF measurements for base station emitters providing mobile wireless telecommunication services is to ensure that all radio channels or resources are occupied during the measurement. This may be verified by knowledge of the system operation or through examination of the signal with a combination of antenna and spectrum analyser. If measurements with all channels or resources occupied are not possible, then an extrapolation procedure should be used. Such procedures are described for various technologies of mobile telecommunication systems in [IEC 62232].

The last step is to calculate the total equivalent electric field strength, E_{TOT} , that will be compared with the exposure limit. It is obtained by the RMS sum of the contribution from each transmission system (numbered by the index k):

$$E_{TOT} = \sqrt{\sum_k E_k^2} \leq E_{\text{lim}}(f)$$

and when different limits are defined for different frequencies:

$$\rho_E = \sqrt{\sum_k \frac{E_k^2}{E_{\text{lim}_k}^2}} \leq 1$$

Uncertainty: when conservative approximations (e.g., full traffic) are made during processing, the field strength resulting from the post-processing is compared with the exposure limit.

If the power density has been evaluated, similar formulas apply and are given in [IEC 62232].

Appendix I

Calculation methods

(This appendix does not form an integral part of this Recommendation.)

I.1 General

This appendix provides guidance in selecting calculation methods to assess compliance with EMF levels. There are several methods useful for determining compliance with exposure limits:

- 1) finite-difference time-domain (FDTD);
- 2) multiple-region finite-difference time-domain (MR/FDTD);
- 3) synthetic model and ray tracing model;
- 4) hybrid ray tracing/FDTD methods; and
- 5) near-field antenna models such as the method of moments (MOM) and the numeric electromagnetic code (NEC).

The selection of the appropriate numerical method depends on the following factors:

- 1) the field zone where the exposure evaluation is required;
- 2) the quantities being evaluated (SAR or reference fields); and
- 3) the topology of the environment where the exposure occurs.

The selection criteria are summarized in Table I.1:

Table I.1 – Selection of numerical techniques

Field zone	Topology	Evaluated quantity	Suitable numerical technique
Near-field	Open	Field	FDTD, MOM
Near-field	Open	SAR	FDTD
Near-field	Closed, multiple scatterers	Field	FDTD, MOM
Near-field	Closed, multiple scatterers	SAR	FDTD, MR/FDTD
Far-field	Open	Field	Ray tracing, MOM
Far-field	Multiple scatterers (complex urban environment)	Field	Ray tracing

More detailed information on numerical techniques can be found in [IEC 62232].

I.2 Methods description

A brief description of the various methods and more information on calculation methods can be found in the following clauses. More details can be found in [IEC 62232].

I.2.1 FDTD

The FDTD method is most useful for exposure assessment near an antenna or in confined locations with a complex scattering environment. The FDTD algorithm is the most widely accepted computational method for SAR modelling [b-Kunz]. The FDTD method offers great flexibility in modelling the heterogeneous structures of anatomical tissues and organs.

The FDTD method can be used to predict field values in complex scattering environments by specifying appropriate boundary conditions or to predict SAR by specifying the dielectric properties and dimensions of the human body and appropriate boundary conditions for closed or open spaces, see [b-Mur].

A sinusoidal waveform is typically used as the excitation source at the antenna feed-point to perform the computations. The signal is allowed to propagate and interact with the objects modelled in the computational domain by means of numerical iterations. The FDTD algorithm iterates the field propagation in both space and time until the field conditions in the computational domain reach a sinusoidal steady state. The total field at selected tissue locations can be computed to determine the SAR. In order to maintain numerical stability for the computational algorithms, the courant condition that provides the minimum relationship for selecting the time and spatial resolutions in the computation must be used. The iteration speed and expected computational errors are related to the parameters used for meeting the courant condition.

I.2.2 MR/FDTD

The MR/FDTD algorithm [b-Johnson] overcomes computational inefficiencies of FDTD for geometries that include extensive sparse regions. In MR/FDTD the problem space is divided into several independent subregions distributed in an otherwise free space. The fields in the subregions are determined with the use of localized FDTD lattices.

I.2.3 Synthetic model

The synthetic model is used to calculate the electric field strength at an evaluation point using a vector sum of a number of small "patches" of the EUT antenna treated as separate sources [b-Altman]. The synthetic model can be used alone for free space exposure evaluation or together with the ray tracing algorithms to take into account the environment (e.g., ground, walls).

This model is applicable in the radiating near-field and far-field regions.

I.2.4 Ray tracing

Ray tracing is useful for the evaluation of fields in large open areas and in urban environments that involve multiple scatterers. A simple two-ray model is used in [ITU-T K.52]. This model is accurate for open unbounded areas over a flat earth. More complex scattering environments that involve reflections from buildings, fluctuations in earth elevations, etc., require complicated multi-ray algorithms. The main disadvantage of ray tracing is that it is essentially a far-field technique. Also, it assumes that the size of the scatterer is large compared to the wavelength. Ray tracing is not suitable for calculation at long wavelengths, where diffraction is important. Ray tracing does not enable calculation of the SAR.

I.2.5 FDTD/ray tracing

The hybrid FDTD/ray tracing technique [b-Bernardi] tries to obtain the advantages of both methods. These methods use ray tracing to evaluate the incident field and FDTD to evaluate the SAR in the body.

I.2.6 MOM

The method of moments (MOM) [b-Harrington] is useful for evaluating the field strength emanating from antennas or other types of thin-wire conductive structures and for computation of the scattered field from thin-wire metallic structures. The use of MOM for computation of scattering from conductive planar surfaces requires that such surfaces be represented by a wire mesh. MOM is useful for near-field and far-field computations. The details of the antenna construction and geometry and the geometry of scattering objects must be known. The MOM is not useful for determining field penetration through dielectric bodies and, therefore, is not suitable for determining SAR. Commercial and non-commercial implementations of MOM are available.

I.3 Other near-field models

The ray tracing algorithms are most useful for exposure sufficiently far from the radiator where the fields reflected from buildings and the unevenness of the terrain are important. In the majority of

telecommunication applications, the field drops below the limit values a few metres from the source. Therefore, an accurate evaluation of the field near the antenna is required. In addition to MOM described in clause I.2.5, there exist several other methods to evaluate the field if the details of the antenna construction and geometry are known. Such methods can also take into account scattering from objects near the antenna.

I.4 Practical problems

The main practical problem in the application of complex computational techniques, such as ray tracing or NEC is that the geometry needs to be specified precisely. In practice, the biggest obstacle to using even simple two-ray models is the lack of adequate information about the antenna and the exposure environment. For example, the available terrain data may have limited resolution. Another example is when the antenna pattern provided by the manufacturer is valid for the far-field region. Near the antenna, the antenna gain may decrease and lobes may shift. One solution for this is to calculate the antenna patterns using MOM if the antenna construction is known.

Appendix II

Identification and measurement of the main source of radiation

(This appendix does not form an integral part of this Recommendation.)

II.1 Treatment of non-conformities – Exceeding the exposure limits

The parameter that determines the compliance, or non-compliance, of a fixed telecommunication installation such as a base station is the level of cumulative exposure which can be measured with broadband or narrow-band meters (or probes) as defined in [IEC 62232].

In cases where, in a certain access zone, the level of cumulative exposure exceeds the maximum limit, the level of received energy of each radiant source should be measured using a narrow-band meter. It is necessary to highlight that when the level of cumulative exposure exceeds the limit (100%), it can be due to two reasons:

- Existence of one or several radiating sources whose levels of received energy exceed the exposure limits corresponding to the operation frequency.
- The level of received energy of each of the involved radiating sources is lower than the exposure limit corresponding to its operation frequency; nevertheless, the combined effect of multiple sources contributes to the level of cumulative exposure being greater than 100%.

Depending on the results of the measurements one of the following cases may apply.

II.1.1 Case 1: Existence of one or several radiant sources whose levels of received energy exceeds the respective exposure limits

By use of a narrow-band meter, it can be identified which radiant sources exceed the exposure limit corresponding to the operation frequency. Once identified, these should be adjusted using mitigation techniques.

II.1.2 Case 2: The level of received energy of each of the radiant sources involved in the measurement is lower than the exposure limit corresponding to its operation frequency; nevertheless, the level of cumulative exposure is greater than 100%

In this case, the level of cumulative exposure is greater than the limit (100%), even when the radiant sources, previously identified in the case 1, emit an electromagnetic field lower than their respective exposure limit. For this reason, it is necessary to select the sources that contribute the greatest energy, depending on the selection approach. Once identified, these should be adjusted using mitigation techniques.

II.2 Confirmation measurement

Following the employment of the mitigation techniques, it should be verified that the level of cumulative exposure does not exceed 100%.

Bibliography

- [b-Altman] Z. Altman, B. Begasse, C. Dale, A. Karwowski, J. Wiart, M.F. Wong, and L. Gattoufi (2002), *Efficient models for base station antennas for human exposure assessment*, IEEE Trans. Electromagn. Compat., Nov 2002, vol.44, pp. 588-592.
- [b-Bernardi] Bernardi P., Cavagnaro M., Pisa S., Piuze E. (2000), *Human Exposure to Radio Base-Station Antennas in Urban Environment*, IEEE Transactions on Microwave Theory and Techniques, Vol. 48, No. 11, pp. 1996-2002.
- [b-CSN EN 50413] CSN EN 50413 (2019), *Basic standard on measurement and calculation procedures for human exposure to electric, magnetic and electromagnetic fields (0 Hz - 300 GHz)*.
- [b-Harrington] Harrington R.F. (1993), *Field Computation by Moment Methods*, Wiley-IEEE Press.
- [b-ICNIRP] ICNIRP (2020), *ICNIRP guidelines for limiting exposure to electromagnetic fields (100 kHz to 300 GHz)*, Health Physics, 118(5), 2020, pp. 483-524.
- [b-IEC PAS 63446] IEC PAS 63446 (2022), *Conversion method of specific absorption rate to absorbed power density for the assessment of human exposure to radio frequency electromagnetic fields from wireless devices in close proximity to the head and body - Frequency range of 6 GHz to 10 GHz*.
- [b-IEC 60050 103] IEC 60050-103:2009, *International Electrotechnical Vocabulary (IEV) - Part 103: Mathematics – Functions*, Section 103-10-10.
- [b-IEC 60050 712] IEC 60050-712:1992, *International Electrotechnical Vocabulary (IEV) - Part 712: Antennas*, Sections 712-02-02 and 712-02-04.
- [b-JCGM 100:2008] JCGM 100 (2008), *Evaluation of measurement data – Guide to the expression of uncertainty in measurement*.
- [b-Johnson] Johnson J.M., Rahmat-Samii Y., (1997), *MR/FDTD: A multiple-region finite-difference – time-domain method*, Microwave and Optical Technology Letters, Vol. 14, No. 2, pp. 101-105.
- [b-Kunz] Kunz K.S., Luebbers R.J., (1993), *The Finite Difference Time Domain Method for Electromagnetics*, CRC Press.
- [b-Mur] Mur G., (1981), *Absorbing Boundary Conditions for the Finite-Difference Approximation of the time-Domain electromagnetic Field Equations*, IEEE Transactions on Electromagnetic Compatibility, Vol. EMC-23, No. 4, pp. 377-382.

SERIES OF ITU-T RECOMMENDATIONS

Series A	Organization of the work of ITU-T
Series D	Tariff and accounting principles and international telecommunication/ICT economic and policy issues
Series E	Overall network operation, telephone service, service operation and human factors
Series F	Non-telephone telecommunication services
Series G	Transmission systems and media, digital systems and networks
Series H	Audiovisual and multimedia systems
Series I	Integrated services digital network
Series J	Cable networks and transmission of television, sound programme and other multimedia signals
Series K	Protection against interference
Series L	Environment and ICTs, climate change, e-waste, energy efficiency; construction, installation and protection of cables and other elements of outside plant
Series M	Telecommunication management, including TMN and network maintenance
Series N	Maintenance: international sound programme and television transmission circuits
Series O	Specifications of measuring equipment
Series P	Telephone transmission quality, telephone installations, local line networks
Series Q	Switching and signalling, and associated measurements and tests
Series R	Telegraph transmission
Series S	Telegraph services terminal equipment
Series T	Terminals for telematic services
Series U	Telegraph switching
Series V	Data communication over the telephone network
Series X	Data networks, open system communications and security
Series Y	Global information infrastructure, Internet protocol aspects, next-generation networks, Internet of Things and smart cities
Series Z	Languages and general software aspects for telecommunication systems